

SHCHUCHINSK, Kazakhstan

WMO: 289840

Lat: 52.9251N

Lon: 70.2038E

Elev: 395

StdP: 96.67

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.2	-29.9	-36.4	0.1	-32.9	-33.0	0.2	-29.7	10.2	-6.5	9.1	-7.2	1.0	300	0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	29.9	17.2	28.1	16.8	26.3	16.3	19.1	26.1	18.3	24.9	17.6	23.8	3.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	12.6	21.1	16.1	12.0	20.6	15.2	11.3	20.0	56.0	26.4	53.4	25.2	51.0	23.6	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	8.4	7.6	-36.4	32.7	2.8	2.1	-38.4	34.2	-40.0	35.4	-41.6	36.6	-43.7	38.1		
(5)			-36.2	20.7	2.9	1.0	-38.3	21.5	-40.0	22.1	-41.7	22.7	-43.8	23.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.1	-16.7	-15.3	-6.8	4.5	11.9	17.0	17.8	16.7	10.7	3.6	-6.0	-13.7
	DBStd	13.84	7.16	6.77	5.94	5.93	5.11	4.28	3.56	4.08	4.94	5.24	7.31	7.14
	HDD10.0	3751	827	708	522	179	40	3	0	2	49	207	481	733
	HDD18.3	6048	1085	941	780	414	206	73	53	82	234	458	731	992
	CDD10.0	858	0	0	0	16	99	214	242	209	70	8	0	0
	CDD18.3	111	0	0	0	0	7	34	36	30	4	0	0	0
	CDH23.3	1257	0	0	0	6	107	395	347	326	75	1	0	0
	CDH26.7	321	0	0	0	0	18	109	84	90	18	0	0	0
(14) Wind	WSAvg	2.8	2.3	2.7	3.0	3.1	3.2	2.9	2.5	2.4	2.5	2.8	3.0	2.8
(15) Precipitation	PrecAvg	350	15	13	17	21	33	41	84	38	23	25	22	18
	PrecMax	515	40	29	32	53	76	83	185	97	50	53	54	52
	PrecMin	191	5	4	3	3	3	1	20	6	1	2	8	4
	PrecStd	91	8	7	10	13	19	23	47	26	12	13	12	10
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.3	0.2	8.0	23.7	29.2	32.0	32.0	31.6	29.2	21.0	8.2	0.6
		MCWB	-1.0	-1.0	3.5	12.1	15.4	17.3	18.3	17.7	14.8	11.8	5.3	-0.4
	2%	DB	-2.4	-1.8	4.2	19.8	26.2	29.5	28.9	29.1	25.3	17.4	5.9	-0.7
		MCWB	-3.1	-2.6	2.0	10.7	14.1	16.8	17.7	17.3	13.7	9.1	4.0	-1.4
	5%	DB	-4.3	-3.4	2.4	17.0	24.1	27.7	26.8	26.9	22.3	14.7	3.7	-2.2
		MCWB	-5.0	-4.1	0.8	9.5	13.2	16.5	17.5	17.1	12.6	7.8	2.2	-3.0
	10%	DB	-6.1	-5.2	1.0	14.2	21.8	25.6	25.1	24.7	19.2	11.5	1.7	-4.1
		MCWB	-6.8	-5.9	-0.2	8.1	12.5	15.9	17.0	16.3	11.5	6.6	0.5	-4.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.8	-0.7	4.4	13.4	17.1	19.6	20.4	19.9	16.6	11.7	6.3	0.0
		MCDB	-0.3	-0.1	7.2	21.1	24.6	27.7	26.6	27.1	26.1	18.6	7.6	0.4
	2%	WB	-3.0	-2.5	2.3	11.6	15.4	18.4	19.3	18.7	14.7	10.1	4.1	-1.3
		MCDB	-2.4	-1.8	3.9	18.1	23.3	26.1	25.9	25.6	22.6	16.0	5.8	-0.8
	5%	WB	-4.9	-4.1	1.0	9.9	14.2	17.6	18.5	17.8	13.4	8.3	2.3	-3.0
		MCDB	-4.3	-3.5	2.2	15.9	21.9	24.9	24.8	24.5	20.8	13.4	3.6	-2.3
	10%	WB	-6.7	-5.8	-0.2	8.4	13.1	16.7	17.8	16.9	12.1	6.9	0.7	-4.9
		MCDB	-6.1	-5.2	1.0	13.9	20.3	23.8	23.2	23.3	17.9	11.2	1.6	-4.2
(35) Mean Daily Temperature Range	MDBR		8.6	9.2	8.9	10.7	12.5	11.7	10.5	11.4	11.2	9.0	6.6	7.8
	5% DB	MCDBR	7.2	7.9	7.9	15.0	15.8	14.5	13.7	14.7	15.8	14.5	7.7	7.6
		MCWBR	7.0	7.4	6.2	8.0	7.1	5.9	5.6	6.1	6.9	8.0	6.0	7.4
	5% WB	MCDBR	7.1	7.3	7.1	14.0	13.8	12.5	11.4	12.6	13.9	12.6	6.9	7.6
		MCWBR	7.0	7.0	5.9	8.2	7.2	5.7	5.4	6.1	6.8	7.6	5.8	7.5
(40) Clear-Sky Solar Irradiance	taub		0.278	0.306	0.361	0.398	0.394	0.399	0.421	0.409	0.357	0.338	0.311	0.278
	taud		2.083	2.055	1.994	2.155	2.242	2.306	2.235	2.250	2.355	2.303	2.155	2.070
	Ebn at Noon		676	775	802	829	854	853	823	811	816	744	628	591
	Edh at Noon		73	104	137	133	129	122	129	121	96	81	69	62
(44) All-Sky Solar Radiation	RadAvg		1.22	2.35	3.81	4.64	5.73	6.08	5.58	4.83	3.53	1.88	0.92	0.81
	RadStd		0.09	0.09	0.39	0.46	0.50	0.47	0.43	0.37	0.24	0.25	0.12	0.11

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page